

ITT320 INTRO TO COMPUTER SECURITY

Chapter 1 : Introduction to Computer Security

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Objectives

- Overview of Computer Security
- The basics of Network
- Understanding security terminology
- Identify the most common dangers to networks
- Find the best approach to network security for your organization
- Use resources available for network security



Introduction to Computer Security

- The growth of the Internet has brought many ways in which Computer and networks can be compromised and data stolen.
- Internet traffic is far more than just humorous YouTube videos or Facebook updates about our vacations.
- Now it is the heart and soul of commerce, both domestic and international.
- There are so much of our business is transacted online, a great deal of personal information is stored in computers that leads to :
  - Information Safeguard
  - Vulnerability to the systems
  - How to ensure these system and data are safe
  - Our information privacy



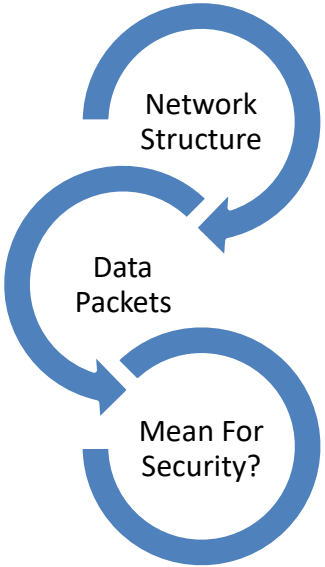
Introduction to Computer Security

- Therefore, in general this course will teach us:
  - What are the current dangers
  - Describes the most common types of attacks on your personal computer and network
  - The security point of view from hackers and security professionals
  - How to secure your computer and network

The Basic of a Network

- In order to manage computer and network security, you will need knowledge about how computer networks operate.

- You need to understand the following:
  - Basic network structure
  - Data packets
  - What does this mean for security?





Basic Network Structure

- Getting two or more computers to communicate and transmit data is called Network. Purpose = Communication.
- How? = Physically connect the computers via a cable that plugs into your computer or wireless connection.
- Both cable and wireless communication relies on a physical device to transmit the data:
  - Wireless Communicaton - network interface card, or NIC
  - Cable Communication - a slot/port for a cable to connect to (RJ-45)
  - The Hub - A hub is a small box-shaped electronic device into which you can plug network cables. Contain 4-24 Ports.
  - The Switch - When a switch receives a packet, it will send that packet only out the port for the computer to which it needs to go.
  - The Router - A router is similar in concept to a hub or switch, as it does relay packets; but we can program/set the router to control how they relay packets.
- Network architecture comprises the format in which these devices are connected.

Basic Network Structure

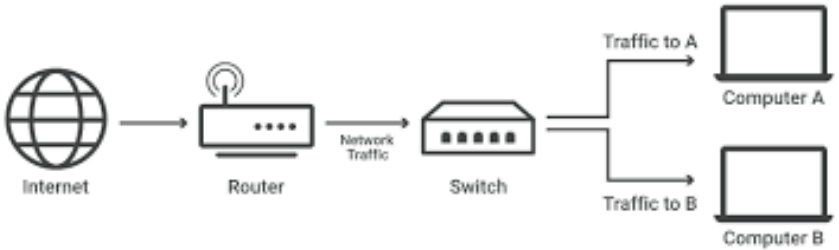
ROUTER







HUB





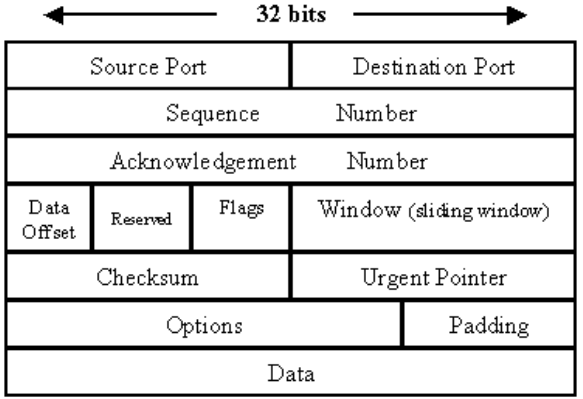
SWITCH

Data Packets

During a Data Transmission, a **Packet** is sent from one machine to another via cable or wireless network.

This is the package that holds the data and transmission information:

- ✓ Ultimately formatted in binary
- ✓ Size ranges between 32 and 65,000 bytes
- ✓ Information included in packets:
- ✓ Source and destination (IP Address) information
- ✓ Packet size (in bytes) and type (e.g. Ethernet)
- ✓ Data and other header information



What Does This Mean for Security?

There are two points of attack:

- ✓ The data packet itself
- ✓ The devices that connect the network

These are also the points of security:

- ✓ Protect the data (encryption)
- ✓ Secure the network devices



Classifications of Threats

Attacks can be classified by function:

1. Intrusion
2. Blocking
3. Malware

Intrusion (Cracking, SE)



Blocking (Dos, DDoS)

Malware (Viruses, Trojan Horses)

Classifications of Threats

Intrusion Types Attack – Any attempt to gain unauthorized access to the system/network

Example of Intrusion Attack:

- ✓ **Security Breaches** – Cracking Password,

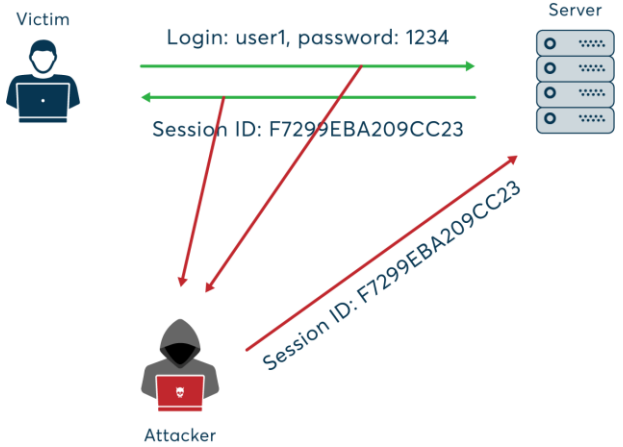
Elevating Privilege, breaking into the server

- ✓ **Web Attacks** – attack that attempts to breach your website,

Via SQL Injection and Cross-Site Scripting *(Cover in Lab Session)*

- ✓ **Session Hijacking** – attacker attempt to take over a session
- ✓ **Social Engineering** - manipulation technique that exploits trust
- ✓ **Insider Threats** – someone who has access to your network -

Steal data / compromise security



Session Hijacking (Source: netsparker.com)

Classifications of Threats

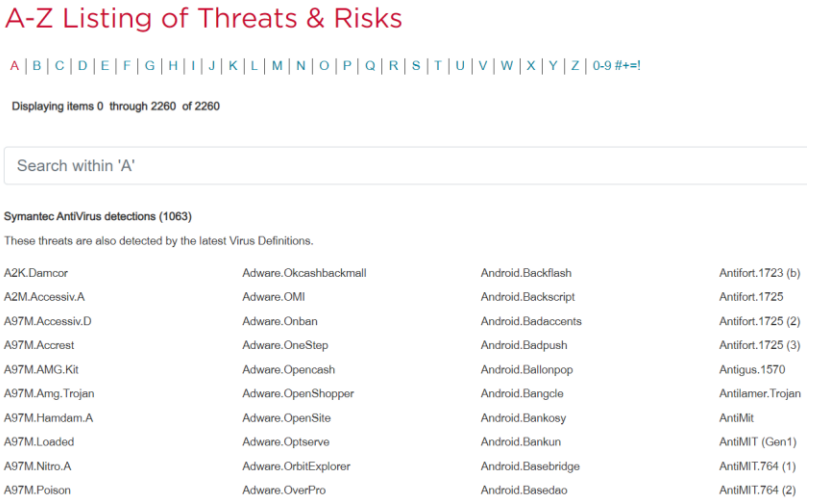
Malware Attack – Stand for Malicious Software, the most prevalent danger to your network

Example of Malware:

- ✓ **Virus**
- ✓ **Worms**
- ✓ **Trojan Horses**
- ✓ **Adware**
- ✓ **Spyware**
- ✓ **Bot**



Ransomware (securityintelligence.com)



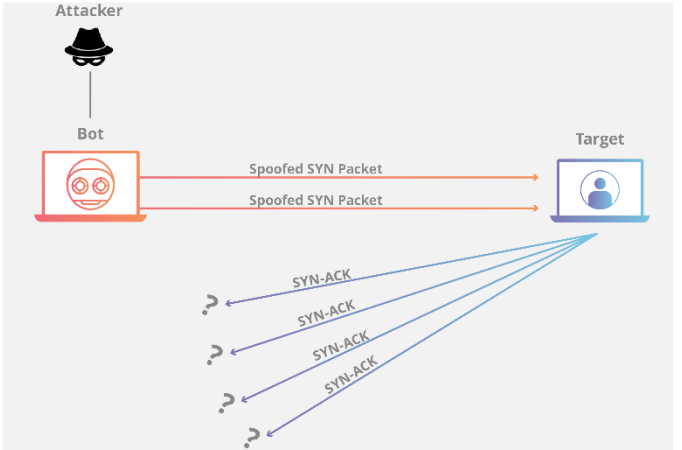
https://www.broadcom.com/support/security-center/a-z

Classifications of Threats

Blocking Attack– Any attempt to shut down a machine or network, making it inaccessible to its intended users

Example of Blocking Attack:

- ✓ **Denial of Services (DoS) –** to prevent legitimate access to your system / website / network (one attacker)
- ✓ **Distributed Denial of Services (DDoS) –** Many attackers / Zombies / Multi-national / Gigabytes of data sent
- ✓ **DNS Poisoning -** compromise a DNS server so that users can be redirected to malicious websites, including phishing websites



DDoS Attack (Source: cloudflare.com)



Classifications of Threats

DOXING

AI Bots / AI Automated Attack

ARE THERE ANY NEW ATTACKS?

Take 5 – Go and Find Out

MEGA DDoS

OSINT

Likely Attacks?

How Seriously Should You Take Threats to Network Security?

"Need a clear picture of the dangers in order to adequately prepare a defense"

Administrators should ask:

- What are the realistic dangers?
- What are the most likely attacks for our network?
- What are some common vulnerabilities?
- What is the likelihood of an attack?



Various Computer Attack (Source: plixer.com)

Likely Attacks?

Threat Assessment

Three factors to consider:

1. Attractiveness of the system (discussed earlier)
2. The nature of the information on the system
3. Traffic to the system (security devices in place)



Likely Attacks?

Assessing Likely Threats to the Network

There are two perspectives in relation to security:

1. First, there is no real threat
2. Second, all hackers are experts and out to break into my network



Likely Attacks?

Assessing Likely Threats to the Network (Cont.)

No real threat:

- ✓ Fosters a laissez-faire attitude toward security
- ✓ Promotes a reactive approach to security
- ✓ Security measures are not put in place until after a breach has occurred
- ✓ This approach must be avoided at all costs



Likely Attacks?

Assessing Likely Threats to the Network (Cont.)

Expert hackers out to get me?

- ✓ Yes, they exist, but not to the extent publicized
- ✓ Lesser skilled hackers are more pervasive
- ✓ These target smaller companies
- ✓ Usually experts seek high profile networks
- ✓ Financial and ideological gain are the targets



Security Terminology

Understanding Security Terminology can be divided to

[Hacking & Security Terminology]



Hacking Terminology (Relate to People & Activity)

Hacker - an expert on a particular system or systems, looking at a system's flaws, and try to exploit to gain access to a system. How the 'EXPLOITING' make them difference :

White Hat Hacker

Black Hat Hacker

Gray Hat Hacker

Security Terminology

Understanding Security Terminology can be divided to Hacking & Security Terminology

How the 'EXPLOITING' make them difference :

1. White Hat Hacker –
   - Find vuln ? will report the flaw to the vendor of that system
   - Can be hired to do Penetration Test
   - E.g Certified Ethical Hacker



Security Terminology

Understanding Security Terminology can be divided to Hacking & Security Terminology

How the 'EXPLOITING' make them difference :



2. Black Hat Hacker –
    1. Gain access to cause harm such as steal data, erase files or deface website
    2. Also referred as crackers

3. Gray Hat Hacker - law-abiding citizen, but in some cases will venture into illegal activities

Security Terminology

Understanding Security Terminology can be divided to Hacking & Security Terminology



Hacking Terminology (Relate to People & Activity)

Script Kiddies :

The person who has no skills, no knowledge and no ethics – they get from crackers or hackers who have the skills.

Ethical Hacking | Penetration Testers:

- ✓ Anyone hired to assess the vulnerabilities of a system
- ✓ Given Permission to assess system vulnerabilities



POWER & GAIN

REVENGE

HACKER'S MOTIVATION ?

Take 5 – Go and Find Out

RECOGNITION

CURIOSITY

Security Terminology

Understanding Security Terminology can be divided to Hacking & Security Terminology

Hacking Terminology (Relate to People & Activity)

Hacktivism:

- ✓ An act of hacking for political reasons
- ✓ Use skills to forward political agenda possibly breaking the law in the process but it will justified because of the political cause



Security Terminology

Understanding Security Terminology can be divided to Hacking & Security Terminology

Security Terminology (Devices & Policies)



Firewall:

A set of related programs, located at a network gateway server, that protects the resources of a private network from users from other networks

Access Control & Authentication:

The management of admission to system and network resources. It grants authenticated users access to specific resources

Security Terminology

Understanding Security Terminology can be divided to Hacking & Security Terminology

Security Terminology (Devices & Policies)

Non-Repudiation:

The assurance that someone cannot deny something.

E.g Digital Signature



Proxy Server, IDS, Auditing and Many More ……

APPROACHING NETWORK SECURITY

Concepts & Approaches

1. CIA Triangles Concepts have Three pillars of security:

   Confidentiality, Integrity, Availability

1. Least Privilege

   User or Services should have the least number of privileges/access required to do her job

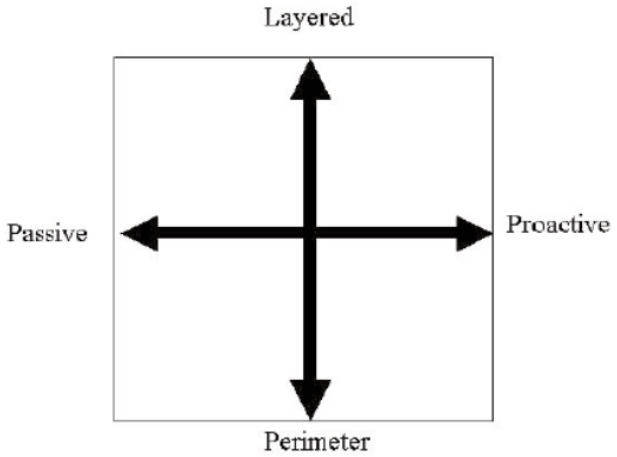


Credit: JamesTyson)

APPROACHING NETWORK SECURITY

Concepts & Approaches

1. Perimeter security approach – Firewall, IDS, IPS
2. Layered security approach - Not only is the perimeter secured, but individual systems such as servers, workstation, routers and hubs
3. Proactive (dynamic) or reactive (passive)
4. Hybrid approach



SECURITY RESOURCES

1. CERT (www.cert.org/)
2. Microsoft Security Advisor (www.microsoft.com/security/default.mspx)
3. F-Secure Corporation (www.f-secure.com/)
4. SANS Institute (www.sans.org/)
5. Security Focus
6. CyberSecurity Malaysia





SUMMARY

Most common dangers to networks:

- ✓ Viruses, worms, trojan horses

Basic security terminology:

- ✓ Hacking terms: deal with people and activities
- ✓ Security terms: deal with devices and policies



SUMMARY

Approaches to securing your network:

- ✓ Proactive vs. reactive
- ✓ Perimeter: focus is on perimeter devices, internal devices are still vulnerable
- ✓ Layered: focus includes both perimeter and individual computers within the network
- ✓ Hybrid: combination of multiple security paradigms



SUMMARY

Resources available for network security:

- ✓ CERT
- ✓ Microsoft Security Advisor
- ✓ F-Secure Corporation
- ✓ SANS institute